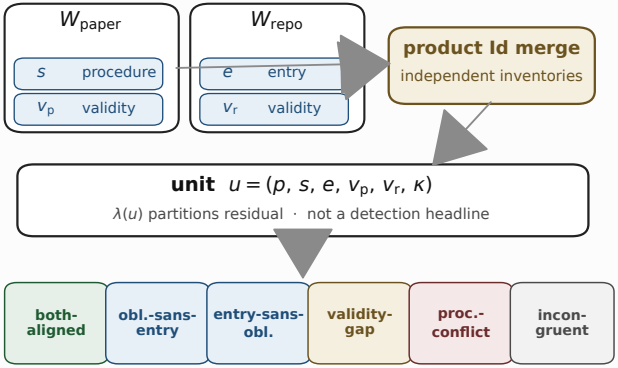
(a) Dual encode → product Id → λ [C1]



C1: product Id before typing · forbids sentence ↔ function host gaps

(b) Coverage fork on H [C1] · qha / band

paper-face gaps ≠ repo-face gaps (can be disjoint)

paper projection
Obl $= \{p : s = 1\}$

qha **obl.-sans-entry** ⇒ **gap**

band **entry-sans-obl.** ⇒ **absent**

repo projection
Ent $= \{p : e = 1\}$

qha **obl.-sans-entry** ⇒ **absent**

band **entry-sans-obl.** ⇒ **gap**

missing ids: paper {qha} ∩ **repo** {band} = ∅

single-face policy cannot be Fl = 0 and Miss = 0 when both omission types nonempty

(c) Operator I: install / refuse / K_H [C2]

λ	action on $H = (P_H, K_H)$	accept / destroy
both-aligned	bind entry; $+K_H$ if shared v	fail → pass; unbind → fail
obl.-sans-entry	write paper steps (no forged e)	fail → pass; drop steps → fail
entry-sans-obl.	REFUSE — $\Delta P_H = \varnothing$	$p \notin$ paper tests
validity-gap	bind + install check → K_H	t_v: 0 → 1; drop check → 0
proc.-conflict	REFUSE silent pick; keep record	H unchanged
incongruent	ignore	—

refusal is success · validity-gap requires $t_v = 1$ **· Props. 2–4**

(d) Contract vs W_{rap} / P_{aper} [C3]

I (typed)
- Fl = 0 (Prop. 2)
- refuse non-obligations
- $t_v = 1$ on val.-gap

W_{rap} / MCP
- Fl = # entry-sans-obl.
- Miss = # obl.-sans-entry
- $t_v = 0$ (no K_H)

P_{aper}
- Fl = 0 but no entry bind
- writes Obl without e
- $t_v = 0$ (Prop. 4)

ablation π: swap omission types → I (Prop. 5: pass sets differ)

Effective
Fl · Miss · t_v

Testable
red→green→ red-on-delete

Fast
calls to green (optional)